Two-Block 4+4→8 Circuit Formation with a μP Readout: Stationary Theory versus Training

Result in one sentence. With the μP-like readout, parent-warm training forms the 4+4→8 circuit in 30 of the theory's 34 H_C cells with near-predicted shares; joint training composes only at $\lambda_0 = 0.03$, and no cell has converged by 250k steps.

Experiment. The $d4/d8$ setting of the theory (exact eight-bit cube; parents $\chi_L = x_1x_2x_3x_4$, $\chi_R = x_5x_6x_7x_8$; target $\chi_8 = \chi_L\chi_R$; $\gamma_4 = \gamma_8 = 2$) in the bias-free two-block residual ReLU network ($N = 1024$, float64; both readouts read h_2), with $\lambda_E = \lambda_2 = \lambda_p = \lambda_8 = \lambda_0$, $\lambda_1 = (\lambda_1/\lambda_2)\lambda_0$ on a 9×8 grid, seed 0, both protocols; SGD with momentum 0.9, no noise, exactly 250k steps. As in the $d2/d4$ runs the readouts have variance $1/N^2$ and rate η/N (μP-like); V_1, V_2 have variance $1/N$ and E variance $1/d$, both at $\eta = 10^{-3}$. Parent-warm cells start from E, V_1, W_{d4} fitted to the parents read from h_1 (50k steps, same optimizer; $R_4 = 0.366$). The previous version of this report used the fan-in readout (variance $1/N$, rate η; now in `fan_in_readout/`); the table repeats its values.

Measurements. As in the Langevin report, with χ_L, χ_R as parents and χ_8 as target: phases, f_C, $R_4 = Q_4/Q_1^{(1)}$ (gate 5/32) and the block-2 4+4→8 shares φ_{out}, φ_{sq}, f_C^{sq}. Fig. 4 adds the block-2 share of the parent amplitude, what block 2's $d8$ producers read ($|w_{8,n}a_n|$-weighted pre-activation power on the parents, on the χ_8 mode made in block 1, a relay of route D_1, or on the singletons, D_2), and $J/\min(F_C, F_D, F_0)$, the objective over the theory's minimal free energy ($F_0 = 3$).

Seed 0, 9×8 grid	Parent-warm μP	Parent-warm fan-in	Joint μP	Joint fan-in	Lower objective μP	Lower objective fan-in
Phase agreement with theory	77.8%	45.8%	54.2%	45.8%	75.0%	45.8%
Gate $H_C \Leftrightarrow R_4 > 5/32$ (active cells)	88.9%	73.1%	45.5%	76.9%	80.0%	76.9%
Mean $\lvert f_C - f_C^{\text{theory}}\rvert$	0.176	0.379	0.273	0.380	0.161	0.379
Cells in H_C (theory: 34)	39	0	4	0	23	0
of which in the theory's H_C region	30	0	4	0	21	0
Cells won (parent-warm / joint)					26/46	43/29
R_4 when the target is first fitted (median)	0.36	0.093	0.23	0.067		
Protocols select different phases	40/72 cells (μP) versus 0/72 cells (fan-in)					

Observations. (i) *Parent-warm reproduces the compositional phase where it is predicted.* In 30 of the 34 theory-H_C cells it ends in H_C, with $f_C = 0.60$–0.94 (theory 0.66–0.88; mean $|\Delta f_C| = 0.09$), $R_4 = 0.19$–0.44 (theory 0.18–0.41) and $\varphi_{\text{out}} = 0.77$–$1.00$, rising with decay as for $d2/d4$ (mean 0.80 at $\lambda_0 = 10^{-4}$, ≥ 0.92 for $\lambda_0 \geq 10^{-3}$). Its block-2 $d8$ producers read the parents (median 98%; relay, singletons and degree-2 leakage $\leq 5\%$, $\leq 13\%$, $\leq 5\%$). At $\lambda_0 = 10^{-4}$, f_C rises with λ_1/λ_2 (0.74 to 0.81) like the theory's (0.68 to 0.88); at $\lambda_0 = 10^{-3}$–10^{-2} it is 0.87–0.94 at every ratio, above the theory at small ratios. (ii) *Deviations sit at the edges.* H_C extends into the theory's H_D region at weak decay (9 cells, $\lambda_1/\lambda_2 = 10$–100, $\lambda_0 \leq 3 \times 10^{-3}$, where the warm start's block-1 parents have not yet decayed; $R_4 < 5/32$ in 5, the gate's only parent-warm failures). At strong decay the target is lost earlier than predicted: at $\lambda_0 = 0.1$ only $\lambda_1/\lambda_2 = 0.1$ stays H_C ($A_8 = 0.68$), $(0.1, 0.3)$ keeps only the parents, $(0.1, 1$–$2)$ collapse, and $(0.03, 5)$ ends at $A_8 = 0.49$, just below the labelling threshold. (iii) *Joint training composes only at* $\lambda_0 = 0.03$ ($\lambda_1/\lambda_2 = 0.1$–$2$: $f_C = 0.67$–0.81 against 0.68–0.84, $\varphi_{\text{out}} = 0.86$–$0.98$) and collapses at $\lambda_0 \geq 0.1$. At $\lambda_0 = 10^{-3}$–10^{-2} it fits the target in 4k–10k steps with $R_4 = 0.19$–0.28, already above the gate, yet its block-2 $d8$ producers read mainly the χ_8 mode built in block 1 (relay, median 58%) and only 34% the parents (H_D, $f_C = 0.15$–0.29); at $\lambda_0 \leq 3 \times 10^{-4}$ they read mainly singletons (60%). The relay, absent from the theory's two route families, rather than accessibility keeps joint training direct; the protocols disagree in 40 of 72 cells, as for $d2/d4$. (iv) *Objectives barely separate the protocols.* Where they disagree inside the theory's H_C region (25 cells) they differ by $\leq 4.1\%$; parent-warm's H_C is lower in 16, joint's H_D in 9 (five at $\lambda_0 = 10^{-4}$), so the lower-objective run is H_C in 21 of the 34 cells. (v) *Nothing has converged.* Every non-suppressed cell still loses 1.7–35% of J per 50k steps. J is carried by the E, V_1, V_2 penalties while the readouts, trained at $\eta/N \approx 10^{-6}$, are still small ($\|w_8\|^2 \leq 0.35$); at $\lambda_0 \leq 3 \times 10^{-4}$ the random initial hidden weights remain ($\|V\|^2 \approx 250$–1040). In parent-warm H_C cells J/F_C falls from 2.4–56 ($\lambda_0 \leq 10^{-3}$) to 0.97–2.85 ($\lambda_0 \geq 3 \times 10^{-3}$; 1.05–1.13 at 0.03): J approaches F_C from above. (vi) *Why the readout matters.* With the fan-in readout the target was fitted within 1k steps at median $R_4 = 0.07$–0.09, and block 2 then relayed D_1; with the μP readout it takes 3k–156k steps and, after pretraining, is fitted through parents that are already accessible. The base-width run ($N_0 = 64$; separate figure) lies between them: target fitted in 1k–27k steps at median $R_4 = 0.17$ (parent-warm) and 0.12 (joint), block 2 relaying D_1 (79–94%), and H_D in both protocols except at $\lambda_0 = 0.1$ (3 and 2 H_C cells). (vii) Readout-free shares are higher ($f_C^{\text{sq}} = 0.69$–0.99 in parent-warm H_C cells; Fig. 2), so the readout favours the block-1 route; block 2 supplies 11–76% of the parent amplitude there, as each 4+4→8 unit also emits the parents, $\sigma(a\chi_L + a\chi_R) = \frac{a}{2}(1 + \chi_L + \chi_R + \chi_L\chi_R)$. Final MSE is set by λ_0, equally for both protocols (Fig. 3).

Limits. One seed; no cell is stationary, so longer training or the $d2/d4$ rate 10^{-2} is the next test of the weak-decay H_C cells and the f_C excess.

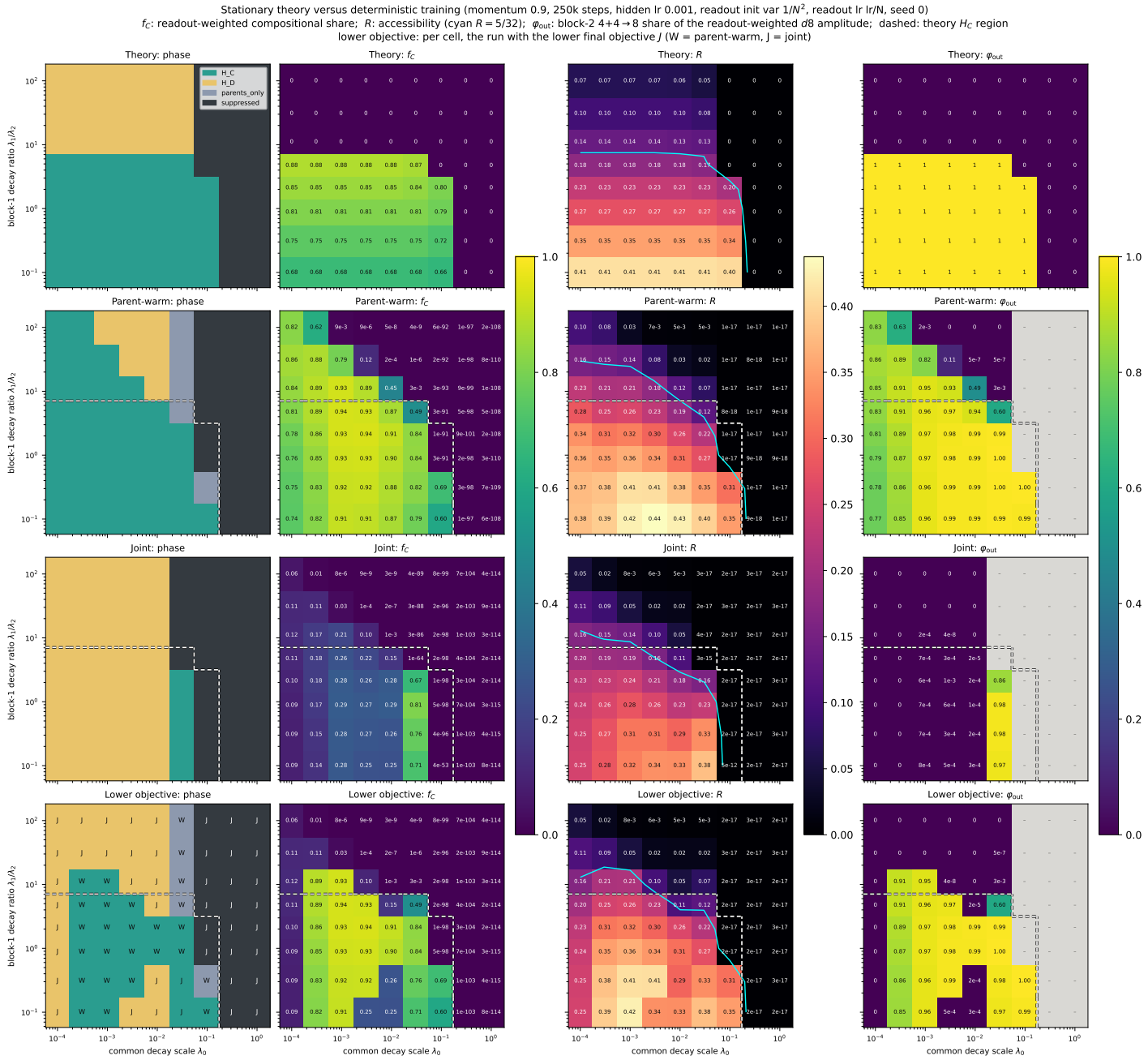


Figure 1: **Theory, parent-warm, joint, and the lower-objective run per cell (rows, top to bottom).** In the bottom row each cell shows whichever of the two runs ended with the lower final objective J (W = parent-warm, J = joint). Columns: phase, readout-weighted compositional share f_C, accessibility R_4 (cyan: $R_4 = 5/32$), and the block-2 4+4→8 share φ_{out} of the readout-weighted $d8$ amplitude. Dashed: theory H_C region. Grey: no target mode produced.

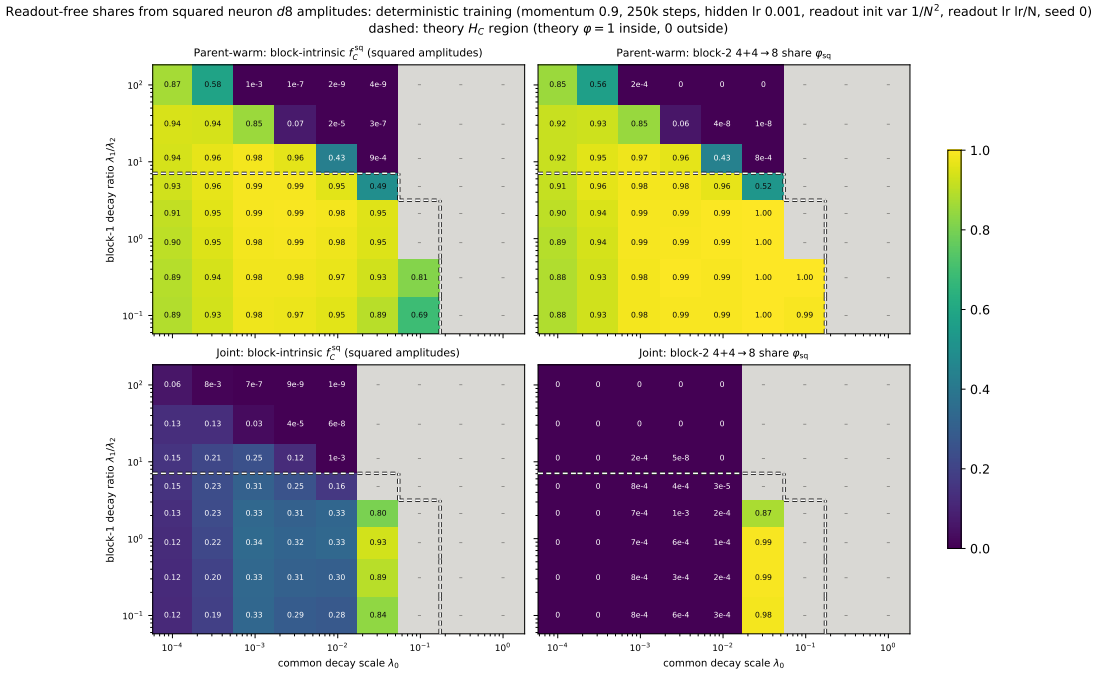


Figure 2: **Readout-free shares** from squared neuron $d8$ amplitudes: f_C^{sq} (left) and φ_{sq} (right); compare with columns 2 and 4 of Fig. 1.

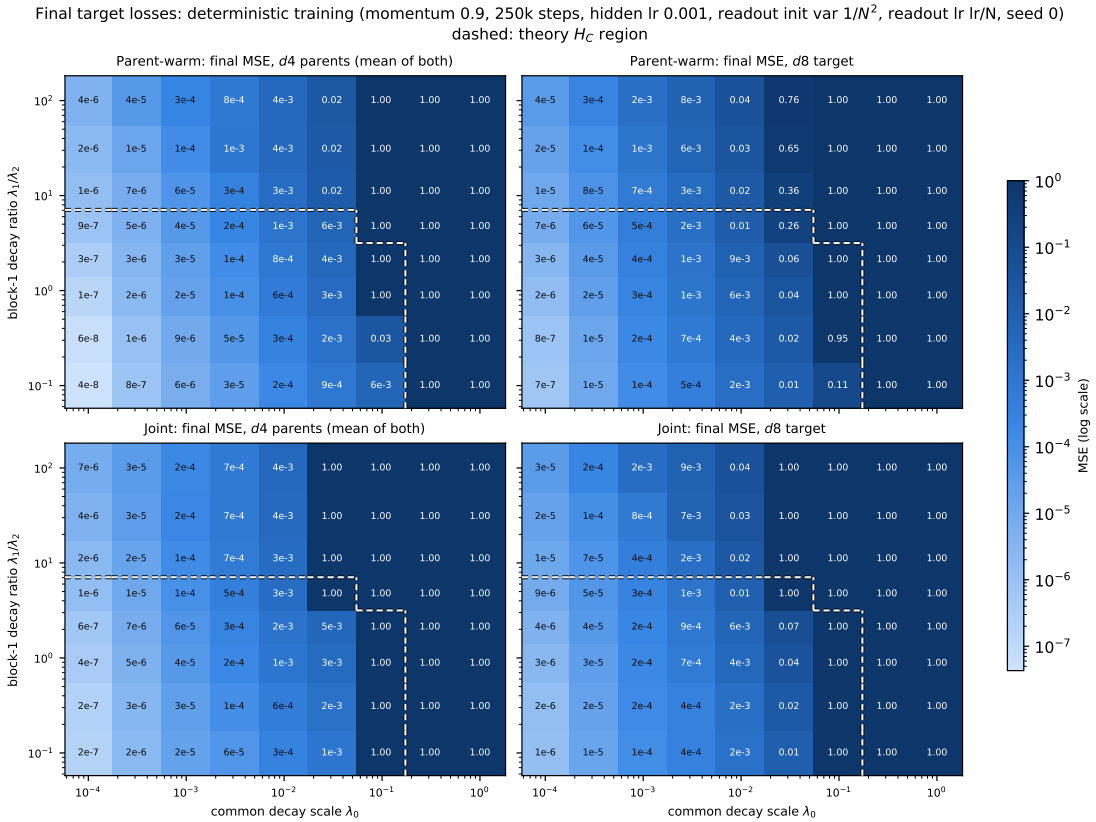


Figure 3: **Final losses:** MSE of the two $d4$ parents (mean) and of the $d8$ target, log scale.

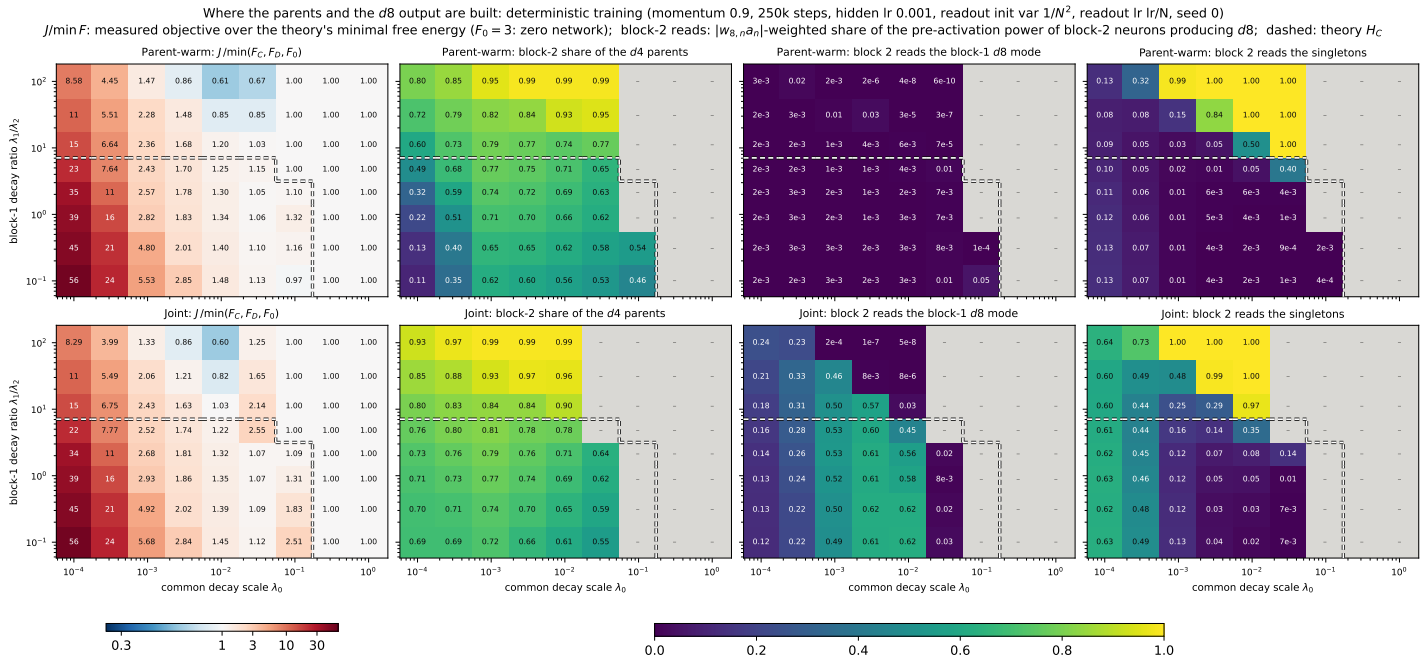


Figure 4: **Where the parents and the** $d8$ **output are built** (rows: parent-warm, joint). Columns: final objective over the theory's minimal free energy (blue: below it; red: above it); block-2 share of the readout-weighted parent amplitude; share of the pre-activation power of block-2 $d8$ producers (weighted by $|w_{8,n}a_n|$) on the χ_8 mode built in block 1 (relay of D_1) and on the eight singletons (direct D_2); the remainder is mostly the parents (4+4→8). Grey: block 2 produces no $d8$.